

A Precision-First Whitespace Restoration Pipeline for Noisy Urdu Tafseer Text

Muhammad Mubashir Hassan

Independent Researcher

Pakistan

mobizaidi123@gmail.com

Abstract

Document conversion and import are a large but under-studied source of real-world noisy text, distinct from the social-media noise usually studied in normalization. We study missing-space artifacts in a low-resource, Arabic-script setting: an Urdu Qur’anic tafseer exported from InPage .inp to .txt and then to JSON/markdown, in whose output we observe fused-token artifacts—adjacent words appearing as a single whitespace-delimited token (e.g., اور قرآن for اور قرآن, “and the Qur’an”). Restoring spaces is hard: Urdu’s cursive Arabic script does not require spaces at word boundaries, and the text interleaves Urdu prose with Qur’anic Arabic, proper names, and markdown, so a careless split can corrupt preservation-sensitive content. We cast this as precision-first normalization and present a confidence-gated pipeline that proposes only space-insertion edits, verifies a character-preservation invariant, and holds protected regions out of automatic correction. On a 1,000-row silver-labeled sample, the high-confidence auto-accept operating point reaches 96.9% strict precision on decisive silver-labeled cases, against 30.5% for an ungated function-word splitter and 39.7% for a frequency-only baseline, with 0 character-altering edits. This suggests that confidence gating is necessary before applying whitespace restoration to preservation-sensitive Urdu–Arabic text. Our labels are model-assisted silver annotations that partly share lexical and corpus-frequency evidence with the pipeline, so we treat the precision figures as provisional rather than as independent estimates, and the derived artifacts can be released for reproducibility.

1 Introduction

Text normalization has been studied largely on user-generated social-media text (Han and Baldwin, 2011; van der Goot et al., 2021), but noise is not only a property of social media. Large volumes

of digitized documents—scanned books, legacy desktop-publishing files, and format migrations—introduce their own systematic, real-world noise during conversion and import. We study one such artifact in a low-resource, Arabic-script setting (Darwish et al., 2021): *missing-space* errors in Urdu, where two adjacent words appear fused into a single token in text produced by a pipeline that converts InPage .inp desktop-publishing files to plain text and then to a JSON/markdown representation. For example, اور قرآن (aur Qur’an, “and the Qur’an”) should be اور قرآن, and کا حق (ka haq, “the right of”) should be کا حق. We observe these fused tokens in the converted representation and attribute them to the InPage-to-text export and the downstream JSON/markdown conversion taken together (§3). Framed for W-NUT, this is multilingual, low-resource lexical normalization on real-world converted text, with the added constraint that the text is *preservation-sensitive*.

Restoring spaces in Urdu is intrinsically hard. Urdu uses a cursive Arabic script in which the visual boundary between words is created by the non-joining behaviour of letters rather than by spaces, so a missing space is often invisible to the eye and cannot be recovered by simple whitespace heuristics. The problem is compounded in tafseer (Qur’anic commentary), where Urdu prose is interleaved with Qur’anic Arabic, hadith quotations, religious terminology, proper names, and markdown formatting. In this mixed Urdu–Arabic setting a careless split can corrupt sensitive or sacred content, so blind correction is unacceptable.

We therefore adopt a *precision-first* stance: it is safer to leave an uncertain spacing error uncorrected than to introduce a wrong split. Our pipeline proposes only space-insertion edits, assigns each a confidence level from corpus-internal lexical evidence, verifies a character-preservation invariant, and holds protected Arabic and markdown regions

out of automatic correction. It *auto-accepts* only high-confidence edits and *defers* or *rejects* the rest.

Contributions.

- We define a preservation-critical whitespace restoration task for mixed Urdu–Arabic tafseer markdown arising from conversion/import artifacts.
- We present a confidence-gated, space-only correction pipeline that prioritizes precision over recall.
- We introduce a character-preservation invariant that verifies no non-whitespace character is modified.
- We evaluate on a 1,000-row silver-labeled sample and show that the high-confidence auto-accept operating point reaches 96.9% precision on decisive silver-labeled cases, while medium/low candidates are unsafe for blind correction; we give an error analysis of why naive function-word splitting fails, and we state explicitly which evidence the silver labels share with the pipeline they evaluate.

2 Related Work

Lexical and noisy-text normalization. Normalizing noisy text has been a central theme of the W-NUT line of research, beginning with lexical normalization of social-media text (Han and Baldwin, 2011) and maturing into multilingual shared tasks such as MultiLexNorm, which normalizes 13 language variants (van der Goot et al., 2021). This work targets user-generated content, where noise stems from informal spelling and abbreviation. We study a complementary source of real-world noise—document conversion and import—which produces systematic, deterministic artifacts (here, fused tokens) rather than stylistic variation.

Urdu word segmentation and space omission. Urdu word segmentation is a long-standing problem because spaces in Urdu are neither necessary nor sufficient to mark word boundaries. Durrani and Hussain (2010) formulate Urdu word segmentation as a joint problem of space insertion and space deletion, and Lehal (2010) specifically targets the space-omission problem in Urdu script. Akram and Hussain (2010) address segmentation as a component of an Urdu OCR system, and

Bin Zia et al. (2018) cast Urdu word segmentation as sequence labeling with conditional random fields. Our task overlaps with this line of work but is deliberately narrower and safety-constrained: we only *insert* spaces, we never delete spaces or alter letters, and we operate on mixed Urdu–Arabic markdown where much of the text must be held out from correction entirely.

OCR and post-conversion correction. Correcting artifacts introduced by digitization is closely related to OCR post-correction, surveyed by Nguyen et al. (2021). Rijhwani et al. (2020) show that OCR post-correction is especially valuable in low-resource settings, where models must exploit limited signal and avoid introducing new errors. Our artifacts arise not from OCR but from an InPage-to-text export and downstream JSON/markdown conversion; the correction is nonetheless a post-conversion normalization problem on noisy text, and it shares the low-resource, error-averse character of that setting. A distinguishing feature of our task is that the text is preservation-critical: much of it (Qur’anic Arabic and quoted material) must be excluded from correction rather than normalized.

Positioning. Prior work addresses Urdu segmentation and OCR correction, but our setting differs because correction is constrained to preservation-safe whitespace insertion in mixed Urdu–Arabic markdown. We do not claim a new segmentation model; we contribute a confidence-gated, character-preserving *operating policy* and an evaluation methodology tailored to preservation-critical text.

3 Data and Task

Corpus. The corpus used in this study is an Urdu Qur’anic tafseer known as *Tafsir al Kashaf*, also referred to in some regions and project metadata as *Tafseer Al-Kashif*. The resulting corpus contains 114 documents (one per surah), 28,072 lines, and 1,581,442 whitespace-delimited tokens. Table 1 summarizes the corpus and the candidates our detector produces from it.

From InPage to the study input. The material reached us in three stages. (i) The Urdu publishers and translators supplied the tafseer as InPage .inp desktop-publishing files. (ii) These files were exported to plain Unicode text using InPage’s built-in text-export function. (iii) The exported text was

Quantity	Value
Documents / surahs	114
Lines	28,072
Tokens	1,581,442
Split candidates	54,139
Protected candidates	16,439
High-confidence candidates	2,134
Medium-confidence candidates	1,572
Low-confidence candidates	50,433
Review sample	1,000

Table 1: Corpus and candidate statistics.

then parsed into a JSON/markdown representation with one document per surah; this converted representation is the input our study operates on. Comparing it against the lexical sequence a reader of the tafseer would expect, we observe fused-token artifacts in the exported and downstream representation: a word pair that should be separated by a space instead appears as a single whitespace-delimited token.

Expected lexical sequence	اور قرآن
Observed converted token	اور قرآن
Proposed restoration	اور قرآن

Because Urdu word boundaries are marked by non-joining letters rather than by spaces, the fused form remains readable and the defect is easy to overlook. We make no claim about how InPage represents or emits spacing internally, and we do not claim that the defect is present in the publishers’ original files. The exact stage at which the whitespace is lost cannot be isolated from the converted artifacts available to us, so we conservatively attribute the noise to the InPage-to-text export and the downstream JSON/markdown conversion taken together rather than to the original publisher source. The task studied here is the recovery of word boundaries in the converted representation, not the reverse engineering of InPage itself.

Availability. Redistribution of the full text depends on the source permissions and is out of scope for this paper. To support reproducibility, the derived candidate records, the silver-labeled evaluation sample, aggregate statistics, and the processing scripts can be released independently of the full corpus text.

Task definition. The input is noisy Urdu markdown. The system may perform exactly one kind of edit: *insert spaces*. The following edits are forbidden: letter changes, letter deletion or insertion,

Merged	Translit.	Restored	Gloss
اور قرآن	aur Qur’an	اور قرآن	and the Qur’an
باغوں میں	baghoṇ meṇ	باغوں میں	in the gardens
کا حق	ka haq	کا حق	the right of
ہیں لیکن	haiṇ lekin	ہیں لیکن	are, but

Table 2: Representative missing-space artifacts with transliteration, the restored (space-inserted) form, and an English gloss.

Arabic normalization, and markdown restructuring. For each suspected merged token the system emits one of three decisions: *auto-accept* (apply the proposed split), *reject* (do not split), or *defer* (hold the token out of automatic correction). Table 2 shows representative merged tokens with transliteration and gloss.

4 Method

Our pipeline has five stages: protected-region detection, candidate detection, confidence assignment, space-only verification, and a decision policy.

Protected-region detection. Because the corpus mixes sacred Arabic with markdown, we first mark regions that must be held out from automatic correction: markdown headings, blockquotes, code fences, tables, links, and images (line level); inline emphasis spans and Qur’anic reference brackets  ...  (span level); and any token bearing Qur’anic vocalization marks or an end-of-ayah symbol. Ordinary Urdu diacritics that are common in prose (such as the superscript alef in تعالیٰ) are deliberately *not* treated as protection signals, to avoid over-flagging normal text.

Candidate detection. We scan for tokens in which a short Urdu function word (e.g., اور, میں, ہے, نے, پر, کے, کی, کا, کو, سے) appears fused to a neighbouring word as a prefix or suffix, and we propose a single space-insertion split.

Corpus-internal lexical evidence. We avoid an external lexicon by using the corpus itself as a frequency lexicon. For a candidate token t that splits into a function word and a remainder r , we use the corpus frequency of t and of r . The seam is chosen to maximize the standalone frequency of r . The guiding intuition is that a genuine missing-space defect is a *rare* fused token whose remainder

Algorithm 1 Per-token correction decision

Require: token t , corpus frequency f , protected flag p
Ensure: decision in {ACCEPT, DEFER, REJECT}

```
1:  $S \leftarrow$  function-word prefix/suffix seams of  $t$ 
2: if  $S = \emptyset$  then return REJECT  $\triangleright$  no candidate
3: end if
4:  $s^* \leftarrow \arg \max_{s \in S} f(\text{remainder}(t, s))$ 
5:  $y \leftarrow \text{INSERTSPACE}(t, s^*)$ 
6: if  $\rho(t) \neq \rho(y)$  then return REJECT  $\triangleright$  fails space-only
7: end if
8:  $c \leftarrow \text{CONFIDENCE}(t, s^*, f, p)$ 
9: if  $c = \text{HIGH}$  and  $\text{rule}(s^*) \notin \text{BAD}$  and  $t \notin \text{OVERRIDE}$  then
10:   return ACCEPT  $y$ 
11: else if  $c = \text{MEDIUM}$  then return DEFER  $y$ 
12: else return REJECT
13: end if
```

is a *frequent* standalone word, whereas a frequent fused token is almost certainly a real single word.

Confidence assignment. Let $f(\cdot)$ be the corpus frequency, t the fused token, and r its remainder after removing the matched function word w . We assign confidence with the following thresholds, applied in order: (i) if $w \in \{\text{وہ، تو}\}$ (frequent word-initial substrings) or the candidate is in a protected region, confidence is *low*; (ii) if $f(t) \geq 8$, the fused form is an established word, so *low*; (iii) a word-final تھی/تھا/نے suffix is capped below *high* (the verb-morphology trap, §6); (iv) otherwise *high* if $f(r) \geq 10$ and $f(t) \leq 3$; *medium* if $f(r) \geq 10$ and $f(t) \leq 7$, or $f(r) \geq 3$ and $f(t) \leq 3$; *low* otherwise. Algorithm 1 summarizes the per-token decision.

Space-only verification. Let $\rho(\cdot)$ denote the function that removes all whitespace from a string. Every proposed split y of a token x must satisfy the character-preservation invariant

$$\rho(x) = \rho(y).$$

If the non-whitespace character sequence is unchanged, then no Urdu or Arabic letter was changed, deleted, or added: the edit is space-only. This invariant certifies character preservation only; it does *not* prove that a split is linguistically correct.

Decision policy. The system auto-accepts a split only when it is high confidence, passes space-only verification, lies outside protected regions, does not come from a known-bad rule, is not on an authoritative override list, and has no competing valid seam. Medium-confidence and otherwise uncertain candidates are deferred; low-confidence can-

didates, known real-word traps, and tokens that would require letter-level correction are rejected.

5 Evaluation

Evaluation sample. We draw a 1,000-row evaluation sample from the 54,139 candidates, stratified across confidence levels, detection rules, protected/unprotected status, and surahs, so that it exercises both safe fixes and dangerous cases rather than only easy examples. All 1,000 rows are labeled (100% coverage).

Labels. Each row receives one label from: `correct_split`, `wrong_split`, `wrong_seam`, `real_word`, `arabic_or_protected`, `proper_name`, `ambiguous`, and `needs_scholar_review`. The first six are *decisive*; `ambiguous` and `needs_scholar_review` denote *deferred* cases that are excluded from precision.

Where the evidence comes from. The silver labels and the pipeline both draw on Urdu lexical and corpus-frequency evidence, so we set out separately what each uses. (A) *Candidate generation (system)*. A closed inventory of short Urdu function words is matched as a prefix or suffix of a token, and the seam that maximizes the remainder’s standalone corpus frequency is proposed (§4). (B) *Confidence features (system)*. The corpus frequencies $f(t)$ and $f(r)$, the protected-region flag, membership of a high-false-positive function-word set, and a word-final verb-morphology suffix test. (C) *Evidence for the silver labels*. The labeling procedure reads the candidate record—token, proposed split, matched function word, $f(t)$, $f(r)$, protected-region flag, and line context—and applies, in order: the authoritative overrides of (D); an inventory of Qur’anic and Islamic proper names; a morphological wrong-seam test based on the enclitic نہیں ; the protected-region flag; the high-false-positive function-word set; a frequency test on the fused token $f(t)$; the word-final نے infinitive test; and finally a frequency test on the remainder $f(r)$. It does *not* read the confidence level or the accept/defer/reject decision that it is being used to evaluate. (D) *Contextual audit*. The high-confidence bucket was additionally inspected in context, and the resulting corrections were recorded as authoritative overrides that take precedence over the heuristic label.

Overlap and residual circularity. Table 3 lists which signals are shared. Corpus frequency, the protected-region flag, the high-false-positive function-word set, and the word-final $\dot{\text{ج}}$ heuristic feed both (B) and (C): the silver evaluation is therefore *not* independent of the features it evaluates. Three properties of the procedure limit, but do not remove, the circularity. First, the labeler is blind to the system’s confidence level and decision, so no candidate is labeled `correct_split` merely because the detector was confident about it. Second, the labels are multi-class rather than a binarized copy of that decision, and the shared signals enter at different thresholds and answer a different question: the labeler calls a split correct only when the remainder is a clearly common standalone word ($f(r) \geq 25$), against the $f(r) \geq 10$ that the system’s high-confidence rule requires. Third, strict precision charges the system for five distinct kinds of error—wrong splits, wrong seams, real words, protected-region tokens, and proper names—so agreement on frequency evidence alone is not sufficient for a row to count as a success.

The residual dependence is directly visible in the high-confidence bucket. Of its 226 decisive rows, all 219 `correct_split` labels were assigned by the remainder-frequency criterion, and all 7 errors were assigned by the contextual audit of (D). The 96.9% figure should therefore be read as saying that a stricter frequency test corroborates the auto-accept decision on 219 of 226 decisive rows, and that contextual inspection identified seven further cases that frequency evidence alone would have endorsed. An error mode shared by the pipeline and the frequency criterion—most plausibly a real Urdu word whose remainder is itself a frequent standalone word—would be scored as a success by both unless the audit caught it. We discuss the consequences in Limitations.

Metrics. We report *strict precision*, the share of decisive rows labeled `correct_split`:

$$\text{strict precision} = \frac{\# \text{correct_split}}{\# \text{decisive rows}}.$$

Decisive rows include `wrong_split`, `wrong_seam`, `real_word`, `arabic_or_protected`, and `proper_name`, so wrong splits, real-word splits, protected-region splits, and proper-name splits all penalize the system. We report strict precision separately for the high, medium, and

Signal	System	Labels
<i>Shared</i>		
Function-word seam inventory	✓	✓
Corpus frequency $f(t)$, $f(r)$	✓	✓
Protected-region flag	✓	✓
High-false-positive function words	✓	✓
Word-final $\dot{\text{ج}}$ morphology	✓	✓
<i>Not shared</i>		
Assigned confidence / decision	✓	—
Proper-name inventory	—	✓
Enclitic seam test (ہیں)	—	✓
In-context audit of high bucket	†	✓

Table 3: Evidence used by the pipeline (*System*) and by the silver-label procedure (*Labels*). The shared rows are the source of the evaluation’s dependence on the system (§5); the frequency thresholds differ between the two ($f(r) \geq 10$ vs. $f(r) \geq 25$). †: the audit corrections are also recorded as the authoritative override list consulted by the deployed decision policy (§4); in the evaluation they act on the labels, where they count as errors against the confidence assignment being measured.

low confidence buckets; high-confidence precision is the main auto-accept operating-point metric. We also report the *deferred rate*, $(\text{ambiguous} + \text{needs_scholar_review})/\text{total}$, and the space-only pass rate from the character-preservation check.

6 Results and Error Analysis

Results. Over all 925 decisive rows the blind strict precision is only 30.5% (282/925): a splitter that accepted every candidate regardless of confidence would be wrong most of the time. This baseline is not the operating point. Table 4 breaks precision down by confidence. At the high-confidence auto-accept operating point, the system achieves 96.9% precision on decisive silver-labeled cases, while medium- and low-confidence candidates fall to 36.0% and 0.0%, respectively. The deferred rate is 7.5% (75/1,000). Table 5 gives the full label distribution; the largest decisive class is `real_word` (37.6%), confirming that a naive splitter would most often damage valid Urdu words.

Baselines and ablation. Table 6 compares our confidence gate against simpler policies on the same sample and labels, reporting precision and coverage (the share of the 282 available correct splits captured). An ungated *function-word splitter* that accepts every candidate reaches only 30.5% precision. A *frequency-only* baseline that accepts a split whenever the remainder is a frequent cor-

Confidence	Rows	Decisive	Correct	Precision
High	266	226	219	96.9%
Medium	209	175	63	36.0%
Low	525	524	0	0.0%
All	1,000	925	282	30.5%

Table 4: Strict precision by confidence bucket on the 1,000-row silver-labeled sample. High confidence is the auto-accept operating point; “All” is the blind-splitter baseline.

Label	Count	Share
correct_split	282	28.2%
wrong_split	25	2.5%
wrong_seam	8	0.8%
real_word	376	37.6%
arabic_or_protected	233	23.3%
proper_name	1	0.1%
ambiguous	75	7.5%
needs_scholar_review	0	0.0%

Table 5: Silver label distribution in the 1,000-row review sample.

pus word ($f(r) \geq 10$) improves precision to 39.7% at full coverage; adding a rare-token condition ($f(t) \leq 3$) raises it to 84.1% but still admits protected-region, verb-morphology, and high-false-positive cases. Our full confidence gate reaches 96.9% precision on decisive silver-labeled cases—57.2 points above the frequency-only baseline and 12.8 above the rare-token variant—at the cost of lower coverage. The progression isolates the contribution of each component: frequency evidence, rarity of the fused token, and the protected-region / rule-blacklist filters. Because the target text is preservation-sensitive, we operate at the high-confidence point. All figures are computed against the same silver labels, so the comparison between policies is internally consistent even though the absolute level is not independently verified (§5).

Space-only safety. Table 7 reports the character-preservation check. All 1,000 proposed splits and all 925 decisive splits are character-preserving: 0 character-altering edits, a 100% pass rate. We stress that this certifies preservation only, not linguistic correctness.

Error analysis. The most error-prone rule is `suffix_fw`: `نے`, with 0% precision on its decisive rows. This is the Urdu *infinitive trap*: a large number of valid Urdu words end in `نے`, including infinitive and inflected verb forms such

Method	Acc.	Corr.	Prec.	Cov.
Function-word splitter (all)	925	282	30.5%	100.0%
Frequency-only, $f(r) \geq 10$	710	282	39.7%	100.0%
+ rare token, $f(t) \leq 3$	271	228	84.1%	80.9%
Confidence-gated (ours)	226	219	96.9%	77.7%

Table 6: Baselines and ablation on the 1,000-row silver-labeled sample (decisive rows). *Acc.*: rows accepted; *Corr.*: correct splits; *Prec.*: precision; *Cov.*: share of the 282 available correct splits captured. Each added constraint—a remainder-frequency threshold, a rare-token condition, and finally the full confidence gate (protected-region exclusion, verb-morphology and high-false-positive rule handling)—trades coverage for precision. Our gate reaches 96.9% precision on decisive silver-labeled cases, 57.2 points above the frequency-only baseline.

Check	Items	Failures	Pass rate
Proposed splits	1,000	0	100%
Decisive splits	925	0	100%

Table 7: Space-only preservation checks ($\rho(x) = \rho(y)$, where ρ removes all whitespace).

as `کرنے` (*karne*), `ہونے` (*hone*), `دینے` (*dene*), `رکھنے` (*rakhne*), `لانے` (*lane*), and `کھینے` (*kahne*). Treating a word-final `نے` as a seam therefore splits real words. The trap is specific to the attachment side: matched as a prefix, `prefix_fw`: `نے` reaches 56.0% precision, which is why the confidence rule caps only *word-final* `نے` below high (§4). The asymmetry is word-specific rather than positional: `prefix_fw`: `کی` reaches 5.5% against 35.0% for `suffix_fw`: `کی`, so it is the rule inventory, not the attachment side alone, that carries the signal. The prefix `تو` (`prefix_fw`: `تو`) is similarly unsafe because it is a frequent word-initial substring, whereas `prefix_fw`: `اور` is far more reliable at 81.3% precision because the conjunction `اور` rarely begins an unrelated single word.

Auditing the high-confidence bucket surfaced frequency-plausible but wrong cases that corpus statistics alone miss: `تیراکی` (*tairaki*, “swimming”) must not be split as `تیراکی`; `بہکے` (*bahke*, “led astray”) is a valid word; `کامو` appears as a cited term/name; `آرامکو` is a proper/company name (Aramco); and `جبرو` is a wrong-seam case whose intended read-

ing is **جبر و** (as in *jabr o tafwid*), not **جب رو**. Two letter-level typo cases were also observed; these are out of scope because the system may only insert spaces, never change letters. These cases share a profile that frequency evidence rewards rather than penalizes: the fused form is rare and the remainder (**کی**, **کے**, **کو**) is among the most frequent words in the corpus, which is exactly the pattern the confidence rule treats as strong evidence of a lost space. They are therefore direct support for the design decision that corpus frequency alone cannot carry the auto-accept decision, and they are the reason the operating point additionally requires the protected-region, rule-blacklist and override filters of §4. Because each of these cases turns on lexical knowledge that no corpus count expresses, they also motivate independent expert-gold annotation in future evaluation.

Towards richer candidate generation. The failures above all originate in the candidate generator, which keys on a small closed inventory of function-word strings and so cannot distinguish a genuinely fused function word from the same letter sequence occurring inside a single word. One signal that is largely orthogonal to both the function-word inventory and the frequency counts is the cursive joining behaviour of Urdu letters: they divide into those that connect to the following letter and those that do not (e.g. **ل**, **ا**, **و**, **و**), and the latter terminate a cursive cluster wherever they occur. Since Urdu word boundaries are realized visually by these cluster breaks rather than by spaces (§3), the distribution of joining and non-joining letters around a candidate seam carries information about where a lost space plausibly fell: a seam inside a connected cluster is orthographically less plausible than one at a cluster break. Such features are cheap to compute from the characters alone and independent of corpus counts, and we expect them to be a useful additional input for ranking or filtering candidate seams. We did not evaluate this here and leave it to future work.

7 Conclusion

We studied missing-space artifacts observed after an Urdu tafseer was exported from InPage .inp to text and converted to JSON/markdown, and we framed their repair as a preservation-critical, whitespace-only normalization task on noisy mixed Urdu–Arabic text. Our evaluation

shows that blind correction is unsafe—strict precision over all decisive candidates is only 30.5%—while a confidence-gated policy that auto-accepts only high-confidence, space-only, unprotected edits reaches 96.9% precision on decisive silver-labeled cases. A character-preservation invariant certifies that every proposed edit is space-only, with 0 character-altering edits across all 1,000 proposed splits. Because the silver labels share lexical and corpus-frequency evidence with the pipeline, that figure describes the behaviour of this operating point on this corpus rather than an independent estimate of generalization. The results nonetheless argue for confidence-gated, character-preserving correction rather than unrestricted splitting in sensitive text. Future work should add independent expert-gold validation, enrich candidate generation with orthographic joining features, extend the method to additional corpora and scripts, and study whether the deferred and rejected candidates can be recovered safely without lowering precision.

Limitations

The evaluation labels are model-assisted *silver* annotations, not independent expert-gold labels; although the high-confidence bucket was additionally inspected in context, the medium and low buckets were not independently verified, so their precision figures should be read as provisional. The study uses a single corpus (one Urdu tafseer), and the method, thresholds, and corpus-frequency lexicon are tuned to it; generalization to other Urdu corpora, genres, or scripts is untested. The system deliberately favors precision over recall, so many true missing-space errors are left uncorrected: deferred and rejected candidates are *not* automatically corrected in the current pipeline. The method handles whitespace restoration only; it does not perform spelling correction, OCR letter-error correction, or Arabic normalization, and tokens needing such fixes are out of scope. Finally, redistribution of the full corpus depends on the source permissions; derived annotations, statistics, and scripts can be shared even where the full text cannot.

Dependence of the evaluation on pipeline features. The silver labels were not generated independently of all the features the system uses. As detailed in §5, lexical, morphological, and corpus-frequency evidence informs both the pipeline’s confidence assignment and the labeling procedure;

the labeler is blind to the system’s confidence level and decision and applies the shared evidence at different thresholds, but the two are correlated rather than independent. In particular, an error mode common to both—a real Urdu word whose remainder happens to be a frequent standalone word—is scored as a success unless the contextual audit catches it, and only the high-confidence bucket was audited. The reported precision may therefore overestimate what an independent expert-gold evaluation would find. We accordingly interpret the 96.9% figure as evidence about the behaviour of the current high-confidence auto-accept operating point on this corpus, not as a fully independent estimate of how the pipeline generalizes. Establishing the latter requires independent expert-gold annotation by qualified Urdu editors, ideally with more than one annotator and an inter-annotator agreement figure; this is the priority for future work, and should precede any application of the pipeline at scale to sensitive text.

Ethical Considerations

The source text was provided in permitted form, in InPage .inp format, by the Urdu publishers and translators who prepared it; the spacing artifacts we correct are observed after the InPage-to-text export and the downstream JSON/markdown conversion, and we make no claim about the publishers’ original files. The material is religious text, and its handling is preservation-sensitive: our system is explicitly constrained so that it never changes, deletes, or inserts any non-whitespace character, and it holds Qur’anic Arabic and other protected regions out of automatic correction. This design is intended to avoid altering sacred or quoted content. Full-text redistribution should follow the applicable permissions; where the full text cannot be shared, the derived annotations, aggregate statistics, and processing scripts can be released to support reproducibility. We make no claim of religious authority and intend the work purely as noisy-text normalization. Because the current labels are silver rather than expert-gold, and because they are not fully independent of the pipeline they evaluate (§5), we recommend expert review before any corrected text derived from this pipeline is published or distributed.

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